MPG No. 33.3.133 $\bar{4}'2'm$ (-4'm2' setting) [Type III, tetragonal] [G tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$G_{xy} = \mathbb{G}_{xy}^{(1)}$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & -G_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$G_{yzx} = 3\mathbb{Q}_{u}^{(1)}$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$G_{yzx} = \mathbb{Q}_{0}^{(1)} - \mathbb{Q}_{u}^{(2)}$
$G_{xyz} = \mathbb{Q}_{0}^{(1)} + 2\mathbb{Q}_{u}^{(2)}$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{yzzx} & 0 & 0 \\ G_{xxxy} & G_{xxxy} & G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$G_{xxxy} = -\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)}$
$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)}$
$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -G_{xxxy} & -G_{xxxy} & -G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$G_{xxxy} = \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_{z}^{\beta(1)}$
$G_{zzxy} = \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_{z}^{\beta(1)}$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}^{(4)}_{xy}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxxy} \\ 0 & 0 & -G_{xxxy} \\ 0 & 0 & 0 \\ 0 & G_{yzzx} & 0 \\ -G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = -2\mathbb{G}^{(5)}_{xy} + \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{yzzx} = -\mathbb{G}^{(5)}_{xy} - \mathbb{Q}^{\beta(2)}_{z}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & -G_{yzzx} & 0 & 0 \\ G_{xyxx} & -G_{xyxx} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = \mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{\beta(3)}_{z}$$
$$G_{xyxx} = -2\mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{\beta(3)}_{z}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} & 0 & 0 & G_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & G_{xxyx} & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & G_{yzxz} & 0 & 0 & G_{yzzx} & 0 & 0 \\ G_{xyxx} & G_{xyxx} & G_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = -\mathbb{G}^{\beta(1)}_{4z} + \mathbb{G}^{(1)}_{xy} + 2\mathbb{G}^{(2)}_{xy} + \mathbb{G}^{(3)}_{xy} - 2\mathbb{G}^{(5)}_{xy} + 2\mathbb{Q}^{\beta(1)}_{z} + \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{xxyx} = -\mathbb{G}^{\beta(1)}_{4z} + \mathbb{G}^{(1)}_{xy} + 2\mathbb{G}^{(2)}_{xy} + \mathbb{G}^{(3)}_{xy} + 2\mathbb{G}^{(5)}_{xy} + 2\mathbb{Q}^{\beta(1)}_{z} - \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{zzxy} = 2\mathbb{G}^{\beta(1)}_{4z} + \mathbb{G}^{(1)}_{xy} + \mathbb{G}^{(3)}_{xy} - 4\mathbb{Q}^{\beta(1)}_{z}$$
$$G_{yzzx} = 2\mathbb{G}^{\beta(1)}_{4z} + \mathbb{G}^{(2)}_{xy} - \mathbb{G}^{(5)}_{xy} - \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{yzxz} = 2\mathbb{G}^{\beta(1)}_{4z} + \mathbb{G}^{(2)}_{xy} + \mathbb{G}^{(5)}_{xy} + \mathbb{Q}^{\beta(2)}_{z}$$
$$G_{xyxx} = -\mathbb{G}^{\beta(1)}_{4z} + \mathbb{G}^{(1)}_{xy} + 2\mathbb{G}^{(2)}_{xy} - \mathbb{G}^{(3)}_{xy} - 2\mathbb{Q}^{\beta(1)}_{z}$$
$$G_{xyzz} = 2\mathbb{G}^{\beta(1)}_{4z} + \mathbb{G}^{(1)}_{xy} - \mathbb{G}^{(3)}_{xy} + 4\mathbb{Q}^{\beta(1)}_{z}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & -G_{yzxz} & 0 & 0 & -G_{yzzx} & 0 & 0 \\ G_{xyxx} & -G_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}^{(4)}_{xy} + \mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{\beta(3)}_{z}$$
$$G_{yzxz} = -2\mathbb{G}^{(4)}_{xy} + \mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{\beta(3)}_{z}$$
$$G_{xyxx} = -2\mathbb{G}^{(6)}_{xy} + \mathbb{Q}^{\beta(3)}_{z}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & G_{xxxy} & 0 \\ G_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzxy} & 0 \\ G_{xxyx} & 0 & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{xyyy} & 0 \\ 0 & 0 & -\frac{G_{xxxy}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} + G_{zzxy} \end{bmatrix}$$

$$G_{xxxy} = -\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)}$$
$$G_{zzxy} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)}$$
$$G_{xxyx} = -\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)}$$
$$G_{yzzx} = 2\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)}$$
$$G_{xyyy} = -\mathbb{G}_{4z}^{\beta(1)} + \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)}$$